Analysis Tutorial 9

Problem 25

(a) To show B is $\mathcal{A} \otimes \mathcal{B}$-measurable is to show $B \in \mathcal{A} \otimes \mathcal{B}$.

Note that $\bigcup_{q \in \mathbb{Q}_+} \{f > q\} \times [0, q]$

$$= \bigcup_{q \in \mathbb{Q}_+} \{\omega \in \Omega : f(\omega) > q\} \times [0, q]$$

$$= \bigcup_{q \in \mathbb{Q}^+} \{\omega \in \Omega : 0 \leq q < f(\omega)\} \times [0, q]$$

$$= \{(\omega, q) \in \Omega \times \overline{\mathbb{R}}_+ : 0 \leq q \leq f(\omega)\} = B$$

Now for $q \in \mathbb{Q}_+$, $\{f > q\} \in \mathcal{A}$, $[0, q] \in \mathcal{B}$.
So $B \in \mathcal{A} \otimes \mathcal{B}$

(b) Let $g : \overline{\mathbb{R}}_+ \to \overline{\mathbb{R}}_+$ be defined by $g(t) = \mu(f > t)$

Note that if $a \leq b$, then $\{f > a\} \supseteq \{f > b\}$.
Since μ is σ-finite, it is monotone.
and so $\mu(f > a) \geq \mu(f > b)$
So $g(a) \geq g(b)$.

To show g is measurable, we show $\forall x \in \overline{\mathbb{R}}_+$, $\{g \leq x\} \in \overline{\mathcal{B}}$.
Notice if $a \in \overline{\mathbb{R}}_+$, $g(a) \leq x$, then $[a, \infty] \subseteq \{g \leq x\}$.
So either $\{g \leq x\} = \overline{\mathbb{R}}_+$ $(g(0) \leq x)$,
$\{g \leq x\} = [a, \infty]$, where $a = \min\{g \leq x\}$
$\{g \leq x\} = \emptyset$ if no such a exists.
But each of these is in $\overline{\mathbb{R}}_+$, so we are done

(c) By Tonelli's Theorem,

$$\int_{\Omega \times \bar{\mathbb{R}}_+} \chi_B \, d(\mu \times \lambda_1)$$

$$= \int_\Omega \underbrace{\int_0^\infty \chi_B(\omega, x)\, dx}_{= \int_0^{f(\omega)} 1\, dx = f(\omega)} \, d\mu(\omega)$$

$$= \int_\Omega f \, d\mu$$

But also $\int_\Omega \int_{\bar{\mathbb{R}}_+} \chi_B \, d(\mu \times \lambda)$

$$= \int_0^\infty \int_\Omega \chi_B(\omega, t)\, d\mu(\omega)\, d\lambda(t)$$

$$= \int_0^\infty \int_{\{f > t\}} 1 \, d\mu(\omega)\, d\lambda(t)$$

$$= \int_0^\infty \mu(f > t)\, d\lambda(t)$$

Problem 26

We know that $\lambda_3(G) = \int_G 1 \, d\lambda_3$

$$= \int_{\mathbb{R}^3} \chi_G \, d\lambda_3$$

$$= \int_{-\infty}^\infty \int_{-\infty}^\infty \int_{-\infty}^\infty \chi_G(x,y,z)\, dx\, dy\, dz$$

$$= \int_{-\infty}^\infty \int_{-\infty}^\infty \underbrace{\int_{\{1-x-y\}} 1 \, dz}_{=0}\, dy\, dx$$

$$= 0$$

Problem 27

We need to find where $\int_{\mathbb{R}} |f|^p \, d\lambda < \infty$:
Fix $\alpha \in \mathbb{R}$. Let $p < \infty$

Then
$$\int_{\mathbb{R}} |f|^p \, d\lambda = \int_1^\infty x^{\alpha p} \, dx$$
$$= \lim_{t \to \infty} \int_1^t x^{\alpha p} \, dx$$

If $\alpha = -\frac{1}{p}$, then $\int_1^t x^{\alpha p} dx = \ln t \to \infty$ as $t \to \infty$.
If $\alpha \neq -\frac{1}{p}$, then $\int_1^t x^{\alpha p} dx = \frac{1}{p\alpha + 1}\left[t^{p\alpha+1} - 1\right]$

Then
$$\lim_{t\to\infty} \int_1^t x^{\alpha p} \, dx < \infty \iff \lim_{t\to\infty} t^{p\alpha+1} < \infty$$
$$\iff p\alpha + 1 \leq 0$$
$$\iff \alpha \leq -\frac{1}{p}$$

We know $\alpha = -1/p$ does not work.
So $\alpha < 1/p$

For $p = \infty$, we need f is essentially bounded. That is, $\exists M > 0$ s.t. $\lambda(|f| > M) = 0$.
But if $\alpha > 0$, $f(x) \to \infty$ as $x \to \infty$ and thus no such M exists.
(for $M \in \mathbb{R}$, let $x \in \mathbb{R}$ s.t. $f(x) > M$.
Then $\lambda(|f| > M) \geq \lambda((f(x), \infty)) = \infty$.)
If $\alpha \leq 0$, then $x^\alpha \leq 1$ and so f is essentially bounded.